We create chemistry

# Safety data sheet



## 1. Substance/preparation and manufacturer/supplier identification

**Product name:**
Accotab 330 EC

Use: crop protection product, herbicide

Manufacturer/supplier:
BASF Vietnam Co. Ltd.
Level 23, Deutsches Haus, 33 Le Duan,
Ben Nghe Ward, Dist. 1, Ho Chi Minh City, Vietnam
Telephone: +84 28 3824 3833
Telefax number: +84 28 3824 3832
E-mail address: minh-triet.thieu@basf.com

Emergency information:
18001703 (Vietnam)
Telefax number: +84 28 3824 3832
International emergency number:
Telephone: +49 180 2273-112

## 2. Hazard identification

Classification of the substance and mixture:
Aspiration hazard: Cat.1
Flammable liquids: Cat.3
Acute toxicity: Cat.5 (oral)
Reproductive toxicity: Cat.2 (unborn child)
Specific target organ toxicity — single exposure: Cat.3 (irritating to respiratory system)
Specific target organ toxicity — repeated exposure (Central nervous system, Kidney, Liver): Cat.2
Hazardous to the aquatic environment - acute: Cat.1
Hazardous to the aquatic environment - chronic: Cat.1
Serious eye damage: Cat.1

Label elements and precautionary statement:

Pictogram:

Signal Word:
Danger

Hazard Statement:

| | |
|---|---|
| H226 | Flammable liquid and vapour. |
| H318 | Causes serious eye damage. |
| H303 | May be harmful if swallowed. |
| H304 | May be fatal if swallowed and enters airways. |
| H335 | May cause respiratory irritation. |
| H361 | Suspected of damaging the unborn child. |
| H373 | May cause damage to organs (Central nervous system, Kidney, Liver) through prolonged or repeated exposure. |
| H400 | Very toxic to aquatic life. |
| H410 | Very toxic to aquatic life with long lasting effects. |

Precautionary Statement:

| | |
|---|---|
| P101 | If medical advice is needed, have product container or label at hand. |
| P102 | Keep out of reach of children. |
| P103 | Read carefully and follow all instructions. |

Precautionary Statements (Prevention):

| | |
|---|---|
| P280 | Wear protective gloves, protective clothing and eye protection or face protection. |
| P271 | Use only outdoors or in a well-ventilated area. |
| P210 | Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking. |
| P260 | Do not breathe dust/gas/mist/vapours. |
| P243 | Take action to prevent static discharges. |
| P201 | Obtain special instructions before use. |
| P202 | Do not handle until all safety precautions have been read and understood. |
| P241 | Use explosion-proof electrical, ventilating and lighting equipment. |
| P240 | Ground and bond container and receiving equipment. |
| P242 | Use non-sparking tools. |

Precautionary Statements (Response):

| | |
|---|---|
| P305 + P351 + P338 | IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing. |
| P310 | Immediately call a POISON CENTER or physician. |
| P308 + P313 | IF exposed or concerned: Get medical attention. |
| P304 + P340 | IF INHALED: Remove person to fresh air and keep comfortable for breathing. |
| P303 + P361 + P353 | IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water or shower. |
| P391 | Collect spillage. |
| P331 | Do NOT induce vomiting. |
| P370 + P378 | In case of fire: Use water spray, dry powder, foam or carbon dioxide for extinction. |

Precautionary Statements (Storage):

| | |
|---|---|
| P233 | Keep container tightly closed. |
| P403 + P235 | Store in a well-ventilated place. Keep cool. |
| P405 | Store locked up. |

Precautionary Statements (Disposal):

| | |
|---|---|
| P501 | Dispose of contents and container to hazardous or special waste collection point. |

Other hazards which do not result in classification:
See section 12 - Results of PBT and vPvB assessment.
If applicable information is provided in this section on other hazards which do not result in classification but which may contribute to the overall hazards of the substance or mixture.

## 3. Composition/information on ingredients

Chemical nature

Substance nature: mixture

crop protection product, herbicide, Emulsifiable concentrate (EC)

**Hazardous ingredients**

pendimethalin

| | |
|---|---|
| Content (W/W): 33.85 % | Acute Tox.: Cat. 5 (oral) |
| CAS Number: 40487-42-1 | Repr.: Cat. 2 (unborn child) |
| | Aquatic Acute: Cat. 1 |
| | Aquatic Chronic: Cat. 1 |
| | M-factor acute: 100 |
| | M-factor chronic: 10 |

xylene

| | |
|---|---|
| Content (W/W): < 60 % | Asp. Tox.: Cat. 1 |
| CAS Number: 1330-20-7 | Flam. Liq.: Cat. 3 |
| | Acute Tox.: Cat. 5 (Inhalation - vapour) |
| | Acute Tox.: Cat. 5 (oral) |
| | Skin Corr./Irrit.: Cat. 2 |
| | Eye Dam./Irrit.: Cat. 2B |
| | STOT SE: Cat. 3 (irr. to respiratory syst.) |
| | STOT RE (Central nervous system, Liver, Kidney): Cat. 2 |
| | Aquatic Acute: Cat. 2 |
| | Aquatic Chronic: Cat. 3 |

(OLIGOMER) 4-Nonylphenol, branched, ethoxylated (> 1 < 2.5 mol EO)

| | |
|---|---|
| Content (W/W): < 3 %<br>CAS Number: 127087-87-0 | Acute Tox.: Cat. 4 (oral)<br>Eye Dam./Irrit.: Cat. 2B<br>Aquatic Acute: Cat. 3<br>Aquatic Chronic: Cat. 3 |

methanol

| | |
|---|---|
| Content (W/W): < 0.5 %<br>CAS Number: 67-56-1 | Flam. Liq.: Cat. 2<br>Acute Tox.: Cat. 3 (Inhalation - vapour)<br>Acute Tox.: Cat. 3 (oral)<br>Acute Tox.: Cat. 3 (dermal)<br>STOT SE (Central nervous system, Optic nerve): Cat. 1 |

## 4. First-Aid Measures

General advice:
First aid personnel should pay attention to their own safety. If the patient is likely to become unconscious, place and transport in stable sideways position (recovery position). Immediately remove contaminated clothing.

If inhaled:
Keep patient calm, remove to fresh air, seek medical attention. Immediately administer a corticosteroid from a controlled/metered dose inhaler.

On skin contact:
Immediately wash thoroughly with plenty of water, apply sterile dressings, consult a skin specialist.

On contact with eyes:
Immediately wash affected eyes for at least 15 minutes under running water with eyelids held open, consult an eye specialist.

On ingestion:
Immediately rinse mouth and then drink 200-300 ml of water, seek medical attention. Do not induce vomiting due to aspiration hazard.

Note to physician:
Symptoms: Information, i.e. additional information on symptoms and effects may be included in the GHS labeling phrases available in Section 2 and in the Toxicological assessments available in Section 11., (Further) symptoms and / or effects are not known so far
Treatment: Treat according to symptoms (decontamination, vital functions), no known specific antidote.

## 5. Fire-Fighting Measures

Suitable extinguishing media:
water spray, carbon dioxide, foam, dry powder

Specific hazards:
carbon monoxide, carbon dioxide, nitrogen oxides, sulfur oxides
The substances/groups of substances mentioned can be released in case of fire.

Special protective equipment:
Wear self-contained breathing apparatus and chemical-protective clothing.

Further information:
Collect contaminated extinguishing water separately, do not allow to reach sewage or effluent systems. Dispose of fire debris and contaminated extinguishing water in accordance with official regulations. In case of fire and/or explosion do not breathe fumes. Keep containers cool by spraying with water if exposed to fire.

## 6. Accidental Release Measures

Personal precautions:
Do not breathe vapour/spray.Use personal protective clothing.Avoid contact with the skin, eyes and clothing.

Environmental precautions:
Do not discharge into the subsoil/soil.Do not discharge into drains/surface waters/groundwater.

Methods for cleaning up or taking up:
For small amounts: Pick up with suitable absorbent material (e.g. sand, sawdust, general-purpose binder, kieselguhr).
For large amounts: Dike spillage. Pump off product.
Dispose of absorbed material in accordance with regulations. Collect waste in suitable containers, which can be labeled and sealed. Clean contaminated floors and objects thoroughly with water and detergents, observing environmental regulations. Wear suitable protective equipment.

## 7. Handling and Storage

Handling

No special measures necessary if stored and handled correctly. Ensure thorough ventilation of stores and work areas. When using do not eat, drink or smoke. Hands and/or face should be washed before breaks and at the end of the shift.

Protection against fire and explosion:
Vapours may form ignitable mixture with air. Prevent electrostatic charge - sources of ignition should be kept well clear - fire extinguishers should be kept handy.

Storage

Segregate from foods and animal feeds.
Further information on storage conditions: Keep away from heat. Protect from direct sunlight.

Storage stability:
Storage duration: 24 Months

Protect from temperatures below: 5 °C
The product can crystallize below the limit temperature.
Protect from temperatures above: 40 °C
Changes in the properties of the product may occur if substance/product is stored above indicated temperature for extended periods of time.

## 8. Exposure controls and personal protection

Components with occupational exposure limits

xylene, 1330-20-7;
STEL value 300 mg/m3 (OEL (VN))
TWA value 100 mg/m3 (OEL (VN))
TWA value 20 ppm (ACGIHTLV)

Personal protective equipment

Respiratory protection:
Suitable respiratory protection for lower concentrations or short-term effect: Combination filter for gases/vapours of organic, inorganic, acid inorganic, alkaline compounds and toxic particles (e. g. EN 14387 Type ABEK-P3)

Hand protection:
Suitable chemical resistant safety gloves (EN ISO 374-1) also with prolonged, direct contact (Recommended: Protective index 6, corresponding > 480 minutes of permeation time according to EN ISO 374-1): E.g. nitrile rubber (0.4 mm), chloroprene rubber (0.5 mm), butyl rubber (0.7 mm) etc.

Eye protection:
Tightly fitting safety goggles (splash goggles) (e.g. EN 166)

Body protection:
Body protection must be chosen depending on activity and possible exposure, e.g. apron, protecting boots, chemical-protection suit (according to EN 14605 in case of splashes or EN ISO 13982 in case of dust).

General safety and hygiene measures:
The statements on personal protective equipment in the instructions for use apply when handling crop-protection agents in final-consumer packing. Wearing of closed work clothing is recommended. Store work clothing separately. Keep away from food, drink and animal feeding stuffs.

## 9. Physical and Chemical Properties

| | |
|---|---|
| Form: | liquid |
| Colour: | red-brown |
| Odour: | aromatic, of xylene |
| Odour threshold: | Not determined due to potential health hazard by inhalation. |
| pH value: | approx. 5 - 7<br>(CIPAC standard water D, 1 %(m),<br>approx. 20 °C) |
| crystallization range: | < -5 °C |
| Boiling range: | approx. 137 - 140 °C<br>Information applies to the solvent. |
| Flash point: | 32 °C |

| | | |
|---|---|---|
| Evaporation rate: | not applicable | |
| Flammability (solid/gas): | not applicable | |
| Lower explosion limit: | 1 %(V) | |
| Upper explosion limit: | 7 %(V) | |
| Ignition temperature: | approx. 375 °C | (Regulation 440/2008/EC, A.15) |
| Thermal decomposition: | 210 °C , 820 kJ/kg (onset temperature) | (DSC (OECD 113)) |
| | Not a substance liable to self-decomposition according to UN transport regulations, class 4.1. | |
| SADT: | > 75 °C | |
| Explosion hazard: | not explosive | (Regulation 440/2008/EC, A.14) |
| Fire promoting properties: | not fire-propagating | (Regulation 440/2008/EC, A.21) |
| Vapour pressure: | approx. 10.6 hPa (25 °C) Information applies to the solvent. | |
| Density: | approx. 0.97 g/cm3 (20 °C) | |
| Relative vapour density (air): | not applicable | |
| Solubility in water: | emulsifiable | |
| Partitioning coefficient n-octanol/water (log Pow): | not applicable | |
| Viscosity, dynamic: | approx. 0.74 mPa.s (25 °C) Information applies to the solvent. | |
| Viscosity, kinematic: | approx. 1 mm2/s (40 °C) | |

Other Information:
If necessary, information on other physical and chemical parameters is indicated in this section.

## 10. Stability and Reactivity

Conditions to avoid:
See SDS section 7 - Handling and storage.

| | |
|---|---|
| Thermal decomposition: | 210 °C, 820 kJ/kg (DSC (OECD 113)) (onset temperature) |
| Thermal decomposition: | Not a substance liable to self-decomposition according to UN transport regulations, class 4.1. |

Substances to avoid:
strong oxidizing agents, strong bases, strong acids

Hazardous reactions:
No hazardous reactions if stored and handled as prescribed/indicated.

Hazardous decomposition products:
No hazardous decomposition products if stored and handled as prescribed/indicated.

Chemical stability:
The product is stable if stored and handled as prescribed/indicated.

Reactivity:
No hazardous reactions if stored and handled as prescribed/indicated.

## 11. Toxicological Information

### Routes of exposure

**Acute oral toxicity**

Experimental/calculated data:
LD50rat (oral): 3,148 mg/kg

**Acute inhalation toxicity**

LC50 rat (by inhalation): > 9.38 mg/l 4 h
An aerosol was tested.

**Acute dermal toxicity**

LD50 rat (dermal): > 2,000 mg/kg
No mortality was observed.

**Assessment of acute toxicity**

Of low toxicity after single ingestion. Virtually nontoxic by inhalation. Virtually nontoxic after a single skin contact. The product has not been tested. The statement has been derived from substances/products of a similar structure or composition.

**Symptoms**

Information, i.e. additional information on symptoms and effects may be included in the GHS labeling phrases available in Section 2 and in the Toxicological assessments available in Section 11.
(Further) symptoms and / or effects are not known so far

### Irritation

Assessment of irritating effects:
May cause severe damage to the eyes. Not irritating to the skin. The product has not been tested. The statement has been derived from substances/products of a similar structure or composition.

Experimental/calculated data:
Skin corrosion/irritation rabbit: non-irritant

Serious eye damage/irritation rabbit: irreversible damage

### Respiratory/Skin sensitization

Assessment of sensitization:
There is no evidence of a skin-sensitizing potential. The product has not been tested. The statement has been derived from substances/products of a similar structure or composition.

Experimental/calculated data:
Draize test human: Non-sensitizing.

### Germ cell mutagenicity

Assessment of mutagenicity:
Mutagenicity tests revealed no genotoxic potential. The product has not been tested. The statement has been derived from the properties of the individual components.

### Carcinogenicity

Assessment of carcinogenicity:
The product has not been tested. The statement has been derived from the properties of the individual components.

Information on: pendimethalin
Assessment of carcinogenicity:
In long-term studies in rats the substance induced thyroid tumors. The effect is caused by an animal specific mechanism that has no human counter part. In long-term studies in mice in which the substance was given by feed, a carcinogenic effect was not observed.

Information on: methanol
Assessment of carcinogenicity:
In long-term studies in rats and mice in which the substance was given by inhalation, a carcinogenic effect was not observed. In long-term animal studies in which the substance was given in the drinking water in high concentrations, a carcinogenic effect was observed. These effects are not relevant to humans at occupational levels of exposure.
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### Reproductive toxicity

Assessment of reproduction toxicity:
The results of animal studies gave no indication of a fertility impairing effect. The product has not been tested. The statement has been derived from the properties of the individual components.

### Developmental toxicity

Assessment of teratogenicity:
The product has not been tested. The statement has been derived from the properties of the individual components.

Information on: pendimethalin
Assessment of teratogenicity:
Indications of possible developmental toxicity/teratogenicity were seen in animal studies.

Information on: methanol
Assessment of teratogenicity:
The results of animal studies gave indication of a developmental toxic/teratogenic effects with high doses.
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### Specific target organ toxicity (single exposure)

Causes temporary irritation of the respiratory tract.

Remarks: The product has not been tested. The statement has been derived from the properties of the individual components.

### Repeated dose toxicity and Specific target organ toxicity (repeated exposure)

Assessment of repeated dose toxicity:
The product has not been tested. The statement has been derived from the properties of the individual components.

Information on: xylene
Assessment of repeated dose toxicity:
Repeated exposure may affect certain organs. Damages the central nerve system. The substance can cause changes in the following organs after repeated exposure to large quantities: Liver Kidney

Information on: pendimethalin
Assessment of repeated dose toxicity:
No substance-specific organtoxicity was observed after repeated administration to animals. Adaptive effects were observed after repeated exposure in animal studies.

Information on: methanol
Assessment of repeated dose toxicity:
The substance may cause blindness after repeated ingestion. The substance may cause blindness after repeated inhalation.
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### Aspiration hazard

The product has not been tested. The statement has been derived from the properties of the individual components.

### Other relevant toxicity information

Misuse can be harmful to health.

## 12. Ecological Information

### Ecotoxicity

Assessment of aquatic toxicity:
Very toxic to aquatic life with long lasting effects.
The product has not been tested. The statement has been derived from the properties of the individual components.

Information on: pendimethalin
Toxicity to fish:
LC50 (96 h) 0.196 mg/l, Oncorhynchus mykiss
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Information on: pendimethalin
Aquatic invertebrates:
EC50 (48 h) 0.147 mg/l, Daphnia magna
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Information on: pendimethalin
Aquatic plants:
EC50 (72 h) 0.00408 mg/l, Selenastrum capricornutum

EC10 (72 h) 0.00157 mg/l, Selenastrum capricornutum
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Information on: pendimethalin
Chronic toxicity to fish:
No observed effect concentration (288 d) 0.0063 mg/l, Pimephales promelas
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Information on: pendimethalin
Chronic toxicity to aquatic invertebrates:
No observed effect concentration (21 d), 0.0173 mg/l, Daphnia magna
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### Mobility

Assessment transport between environmental compartments:
The product has not been tested. The statement has been derived from the properties of the individual components.

Information on: pendimethalin
Assessment transport between environmental compartments:
The substance will slowly evaporate into the atmosphere from the water surface.
Following exposure to soil, adsorption to solid soil particles is probable, therefore contamination of groundwater is not expected.
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### Persistence and degradability

Assessment biodegradation and elimination (H2O):
The product has not been tested. The statement has been derived from the properties of the individual components.

Information on: pendimethalin
Assessment biodegradation and elimination (H2O):
Not readily biodegradable (by OECD criteria).
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### Bioaccumulation potential

Assessment bioaccumulation potential:
The product has not been tested. The statement has been derived from the properties of the individual components.

Information on: pendimethalin
Bioaccumulation potential:
Bioconcentration factor: 3,300

Based on a weight of evidence, the compound will not bioaccumulate.
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### Additional information

Other ecotoxicological advice:
Do not discharge product into the environment without control.

## 13. Disposal Considerations

Must be sent to a suitable incineration plant, observing local regulations.

Contaminated packaging:
Contaminated packaging should be emptied as far as possible and disposed of in the same manner as the substance/product.

## 14. Transport Information

**Domestic transport:**

| | |
|---|---|
| UN number or ID number: | UN 1993 |
| UN proper shipping name: | FLAMMABLE LIQUID, N.O.S. (XYLENE, PENDIMETHALIN) |
| Transport hazard class(es): | 3, EHSM |
| Packing group: | III |
| Environmental hazards: | yes |
| Special precautions for user: | None known |

**Sea transport**
IMDG

| | |
|---|---|
| UN number or ID number: | UN 1993 |
| UN proper shipping name: | FLAMMABLE LIQUID, N.O.S. (XYLENE, PENDIMETHALIN) |
| Transport hazard class(es): | 3, EHSM |
| Packing group: | III |
| Environmental hazards: | yes<br>Marine pollutant: YES |
| Special precautions for user: | EmS: F-E; S-E |

**Air transport**
IATA/ICAO

| | |
|---|---|
| UN number or ID number: | UN 1993 |
| UN proper shipping name: | FLAMMABLE LIQUID, N.O.S. (XYLENE, PENDIMETHALIN) |
| Transport hazard class(es): | 3 |
| Packing group: | III |
| Environmental hazards: | No Mark as dangerous for the environment is needed |
| Special precautions for user: | None known |

## 15. Regulatory Information

**Other regulations**

The SDS is composed base on the Global Harmonization Globally Harmonized System of Classification and Labeling of Chemicals and applicable local chemical regulations as following :
- The Chemical Law.
- The decree No. 113/2017/ND-CP, Circulars and their replacement (if yes).
- The decree 42/2020/ND-CP guiding on dangerous goods (DG) list, transport and licensing of DG by road and inland waterway.
- Other applicable local regulations.

In case there is a replacement of the above decrees and their circulars, we will update all relevant data accordingly.

To avoid risks to man and the environment, comply with the instructions for use.

## 16. Other Information

Vertical lines in the left hand margin indicate an amendment from the previous version.

The data contained in this safety data sheet are based on our current knowledge and experience and describe the product only with regard to safety requirements. This safety data sheet is neither a Certificate of Analysis (CoA) nor technical data sheet and shall not be mistaken for a specification agreement. Identified uses in this safety data sheet do neither represent an agreement on the corresponding contractual quality of the substance/mixture nor a contractually designated use. It is the responsibility of the recipient of the product to ensure any proprietary rights and existing laws and legislation are observed.